

Ans Sheet: CBSE Class 10 Annual Exam - Vikram Malhotra (Roll: CBSE-10-103)**Q1: Ohm's Law and Resistance of a Conductor**

Ohm's Law: At constant temp, current I is proportional to Voltage V ($V = I \cdot R$).

Factors affecting resistance:

1. Length: Resistance increases with length of wire ($R \propto L$).

2. Area of cross section: $R \propto \frac{1}{A}$ (thick wire has LESS resistance, thin wire has more)

3. Formula: $R = \rho \cdot \left(\frac{L}{A}\right)$

^ $R = \rho \cdot (L / A)$ where ρ = resistivity of material.

4. Temperature: R increases with rise in temperature for metals.

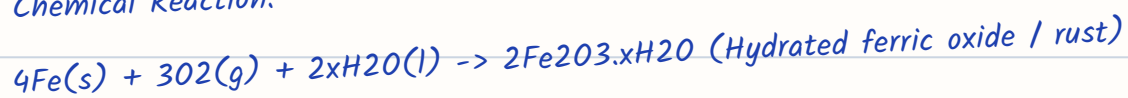
Q2: Rusting of Iron (Corrosion)

Rusting is slow chemical reaction forming reddish brown powder.

~~Rusting happens in dry air alone~~

^ Rusting requires BOTH Oxygen (O_2) and Water/moisture (H_2O)!

Chemical Reaction:



Prevention:

1. Galvanisation: Coating zinc layer on iron sheets.

2. Painting: Applying oil paint to stop contact with air & rain.

Q3: Neutralization Reaction & Antacids

When acid reacts with base to form salt and water.



Antacids: Stomach produces excess HCl causing burning pain & acidity.

Antacids are mild basic substances like Milk of Magnesia [$Mg(OH)_2$].

They neutralize the extra acid giving fast relief.

